

QUESTION 1**(10 MARKS)**

The following code has errors.

```
public static void mainProgram(String[] args) {  
    System.out.println("Found the errors in the triangle");  
    System.out.println(" /\");  
    System.out.println("/  \");  
    System.out.println("----");  
  
    Scanner keybrd = new Scanner(System.in);  
    system.out.println("Enter an integer value one");  
    int value1 = keyboard.nextDouble();  
    System.out.println("The value entered was " + value1);  
  
    System.out.println("Enter another integer value two");  
    int value1 = keyboard.nextInt();  
    System.out.println("The value entered was " + value2);  
    sum = value1 + value2;  
    System.out.println("The sum of the two values are " + sum);  
}
```

Required:

Rewrite the code in your answer books, and at the same time fix any errors. You will lose marks if you change any code that is correct.

QUESTION 2**(10 MARKS)**

Rewrite the following switch statement into an if...else statement.

```
Scanner keyboard = new Scanner(System.in);  
System.out.println("Enter an integer value greater than zero");  
int x = keyboard.nextInt();  
switch(x) {  
    case 1:  
        System.out.println("Value entered was one");  
        break;  
    case 2:  
    case 3:  
        System.out.println("Value entered was a two or a three");  
        break;  
    default:
```

```

        System.out.println("Value entered was greater than three");
    }

```

QUESTION 3 (10 MARKS)

Write if statements for the following sentences. Assume that x and y are defined and have an appropriate value.

3.1. If x is smaller than y, output the message: (3)

Value x is smaller than y

3.2 If x is greater than or equal to y, output the following message: (3)

Value x is greater or equal than y

3..3 If x equals 4 and y equals 6, output the following message: (4)

x equals 4 and y equals 6

QUESTION 4 (5 MARKS)

Change the variables names in the following program code to camel casing. Follow the JAVA accepted norms for variable names.

```

String camelcase = "All variables names in java use camel casing";
double AverageMark = 50.5;
int numberOfstudents = 10;
String listfNames = "Give me the list";
boolean isThisthecorrectCase = false;

```

QUESTION 5 (21 MARKS)

5.1 Write JAVA statements that calculate VAT (Value Added Tax) for any price entered. (7)

You must declare all the variables.

Your program must:

1. Display a message asking the user to enter a price.
2. Read the price as input to your program. Price is a decimal number.
3. Calculate the VAT on the price. VAT is 15%.
4. Display the VAT amount on the screen with an appropriate message.

5.2 Write JAVA statements that calculate a speeding fine for a vehicle that exceeds the speed limit. Declare all the required variables. (15)

The program must:

1. Ask the user to input the speed limit. Read the speed limit as input.

2. Ask the user to input the speed of the vehicle. Read the vehicle speed as input.
3. If the speed of the vehicle is more than the speed limit, calculate and display the fine. The fine is calculated as follows: (speed – speed limit) * 20%.
4. If the vehicle speed is less than speed limit, display an appropriate message.

QUESTION 6

(14 MARKS)

Consider each of the following program extracts and give the answer as asked.

6.1 `int a = 11;` (6)
 `int b = 3;`
 `int c = a % b;`
 `a = a + b;`
 `b = a - b;`

What is value of **a**, **b** and **c** after execution of these statements??

6.2 `int a, b, c, d;` (4)
 `a = 11;`
 `b = 12;`
 `c = 13;`
 `d = 14;`
 `a = b;`
 `b = c;`
 `c = d;`
 `d = a;`

What is value of **a**, **b**, **c** and **d** after execution of these statements?

6.3 `String x, y, z, w;` (4)
 `x = "cape";`
 `y = "town";`
 `z = x + "lovely" + y;`
 `w = 1 + 2 + z + 3 + 4;`

What is value of **z** and **w** after execution of these statements?

*****END OF PAPER*****